Rohan Malhotra

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Education

New York University, Courant Institute — **Aug 2024 - May 2027**
B.A., Computer Science; Math Minor — *New York, NY*
- **GPA:** 3.7/4.0 | Accelerated 3-Year Graduate

Experience

DRW — **May 2026 - Aug 2026**
Applications Developer Intern (Via IBM) — *Dallas, TX*
- Shipped custom financial forecasting implementation including rolling 13-day cash forecasts, SOFR loan calculations, interest accrual models, financial close reports, and XGBoost regression cashflow model in Oracle EPM Planning
- Integrated banking APIs via Oracle Integration Cloud (OIC), building JSON data pipelines to source transaction data into ERP Financials and EPM planning applications. Migrating old Excel sheets onto Oracle Cloud.
- Fine-tuned an open-source 32B language model (Qwen-Coder32B) on Oracle EPM implementation assets, increasing form, business rule, and report creation accuracy from 36.7% to 95.0%.
- Deployed a Text to JSON EPM AI, self-hosted on-prem behind a private encrypted network, integrating RAG, MCP tools, custom agents, and confidence scoring. Using EPM Automate and Oracle EPM API's.

IBM — **Jun 2026 - Aug 2026**
Robotics Engineer Intern — *Dallas, TX*
- Built an end-to-end computer vision pipeline for Boston Dynamics Spot using YOLO11 models trained on 898 manually labeled images, achieving ~99.4% precision, 99.4% recall, and 99.5% mAP@50.
- Integrated real-time inference with the Spot SDK using Python, OpenCV, gRPC, multithreading, and lock-free inference queues; implemented autonomous dog toy finder using the OpenCV & YOLO pipeline.
- Presented live autonomous retrieval demos at IBM DevCon and open-sourced model weights, training pipelines, and deployment documentation. Demos adopted by IBM sales and client success teams for live client engagements.

Kalshi — **Jan 2026 - May 2026**
Quantitative Developer — *New York, NY*
- Co-authored a research paper modeling job-loss risk as a macro-sensitive Poisson hazard process with jump-diffusion income dynamics, applying minimum-variance hedge ratios and Monte Carlo simulation to quantify tail-risk reduction from Kalshi contracts.
- Built a full-stack hedging recommendation engine (Next.js/React + Python) that runs the hazard/Monte Carlo pipeline from user inputs and returns suggested contract sizing, upfront cost, and tail-risk comparisons. Built C++ tools for implied probability conversion, hedge calculations, and real-time market risk analysis.
- Integrated FRED and BLS macroeconomic data APIs into the simulation pipeline; containerized the Node + Python runtime with Docker and deployed to production on Render.

Research

- Hume Center for National Security and Technology: Contributed to design, build, and testing of ContentCube, a CubeSat launched to the ISS and deployed into low Earth orbit; programmed imaging and signal-processing tests in C for autonomous in-orbit photography.
- Reddit Sentiment Data for Financial Applications (hdl.handle.net/10919/124730): NLP and regression models in Python quantifying r/wallstreetbets sentiment impact on equity returns.

Technical Skills

- **Languages**: Python, Java, C/C++, SQL, Groovy, R, MATLAB
- **Technologies**: Git , Linux, Bash, Docker, AWS, PyTorch, LoRA fine-tuning, RAG, NumPy, Pandas, SciPy
- **Financial Tools**: Excel Macros, Oracle EPM Cloud, EPM Automate, Financial Modeling, Monte Carlo